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The Wattenberg Story – Birth of Massive Hydraulic Fracturing

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Abstract

This is a bit of a history lesson, fully describing the development of the Wattenberg field. The successful development of this field with massive hydraulic fracturing led to the opening of the DJ Basin as an important oil/gas producing province. Following the Wattenberg success the basin eventually reached a total gas production of 500 MMscf/day. Wattenberg itself became the 7th largest gas field in N. America producing 194 BCF/year in 2003.

The field was discovered by Amoco Production Co, in 1970 with the drilling of seven exploration and delineation wells. The wells were fractured with typical treatments for that time (20 to 40,000 lbs of 20/40 mesh white sand using a linear gel). The wells responded very well, and after only a short flow period were shut-in as land was leased and production facilities prepared. When the wells were reopened to flow, the surprising result was a rapid production decline with liquid loading killing the wells within a few months. It was the industry's first introduction to the transient flow phenomena of fractured wells in very low permeability formations (on the order of 0.005 mD in this case).

Larger treatments were attempted with very beneficial results, these were reported in the only paper discussing this field. What the paper did not report was that 2 out of 3 of these larger treatments failed due to early total screen-outs. This led to a series of "spectacular failures" (to borrow a term coined by Martin Rylance in a recent Wed(NSI)day presentation) as different procedures were trialed (higher rate, higher viscosity, more pad, fluid loss additives, ...) – all leading to worse results. Basically, operations went from the successful placement of 300,000 lbs of sand to an inability to place even 100,000 lbs !

This paper will review this history and describe how understanding the subsurface geomechanics led to radical changes in treatment design ultimately opening the DJ Basin to production.

Temporal Setting

In the 1950's and 60's, natural gas demand increased exponentially in North America – primarily driven by a vast increase in pipeline delivery systems. [Figure 1](#) provides a chart of the expansion of natural gas use from the 1950's to 2023. This brought attention to vast reserves of natural gas held in very poor quality formations in the western U.S. Primarily in the states of New Mexico, Colorado, and Wyoming. Hydraulic fracturing, first applied commercially in 1949 was routinely used and multiple attempts at large volume fracturing

treatments were trialed by multiple operators. In particular, the authors are familiar with treatments in the Frontier formation in Wyoming, the Mesa Verde formation in Colorado and Wyoming, and the Picture Cliffs formation in New Mexico all unsuccessful. Generally, these attempts were all commercially unsuccessful. There were probably several reasons for this lack of success. These might include geology unsuited to the creation of long fractures, low proppant concentrations seldom exceeding 3 or 4 ppg, poor fluids as the early viscous cross-link gels used 50 to 60 pounds of polymer (per thousand gallons of water) with minimal concentrations of chemical breaker thus leaving massive gel damage in the fractures. Similar large volume viscous fluid fracture treatments were also initially trialed in Wattenberg with a similar lack of success. In the case of the Wattenberg treatments, no chemical breaker was included as it was felt that the 260° F temperature would degrade the fluid.

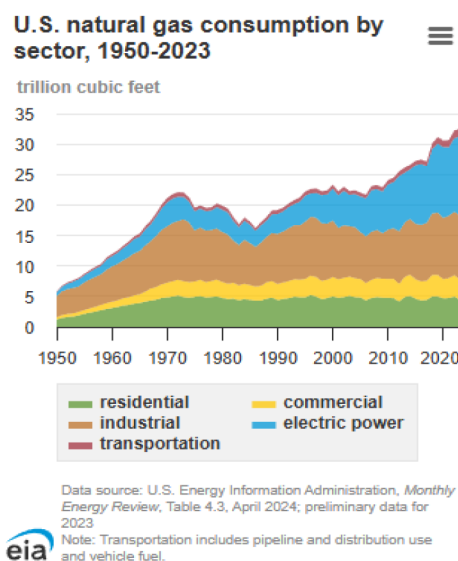


Figure 1—Natural Gas Consumption in North America.

Also, nuclear detonations were trialed (though NOT in the Wattenberg Field) as part of Operation Plowshare which searched for peaceful uses of atomic explosions. Three separate tests were conducted beginning with the Gasbuggy test in 1967 in New Mexico (Figure 2) followed by the Rulison test (1969) and the Rio Blanco test (1973) both in Colorado. All three tests did increase production, but at nowhere near commercial rates. In addition, the radioactive gas was rather problematic. In 2011, the Department of Energy concluded "It was estimated that even after 25 years of gas production of all the natural gas deemed recoverable, that only 15 to 40 percent of the investment could be recovered." (American Oil & Gas Historical Society).



Figure 2—Operation Gasbuggy, 1967.

Wattenberg

The field was discovered in 1970 by Amoco Production Co. and produced from the J-Sand, a large deltaic deposit and member of the Dakota Group. The field is located in the Denver Basin, north of Denver and initially covered about 650,000 acres (primarily in Weld county) – it has since expanded greatly due to the application of horizontal drilling and multiple fracturing stages using the "water fracs" approach.

The field was produced from a deltaic sand formation, the J-Sandstone. (Figure 3). A relatively simple geology, a sand bounded by shale above/below. The formation was naturally under-pressured (about 2,700 psi at 8,000 feet) with reservoir pressure tracked to outcrops to the East in the state of Nebraska.

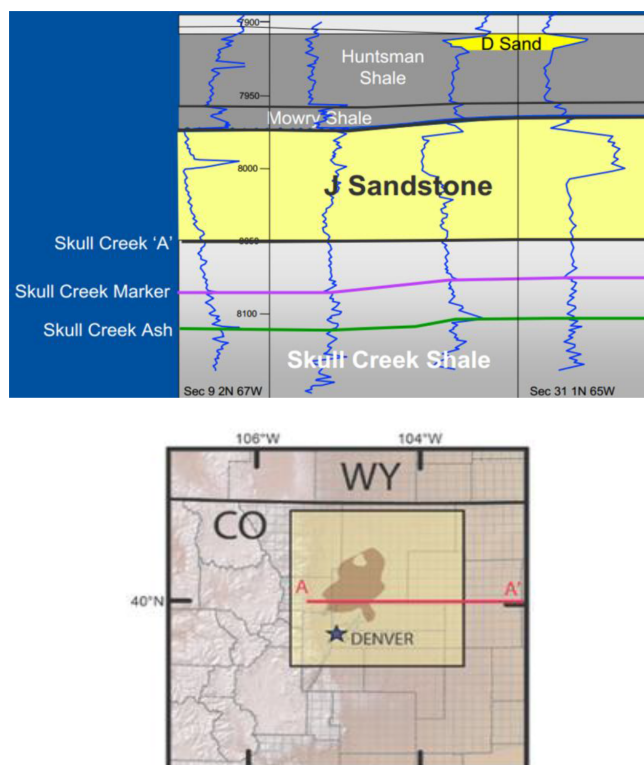


Figure 3—Geologic and Geographical Settings for the Wattenberg Field.

Thus, this was essentially a natural depletion process. While perhaps not immediately desirable from a reservoir viewpoint, this of course resulted in a large minimum in situ stress contrast between the sand and the shale – an ideal environment for hydraulic fracturing. In addition, the field was a gas that was rich in condensate, with a high yield. This aided in the overall economics but also became critical to the use of condensate-emulsion as a fracturing fluid. This became a critical element in the initial success of massive fracturing. Despite this "ideal" environment, initial attempts at large volume fracturing treatments were far from successful as discussed in more detail in sections below.

Subsequent to the initial development of the Wattenberg Field, the DJ Basin has become a significant source of unconventional hydrocarbon reserves known for producing natural gas and condensate from multiple Cretaceous formations including the Muddy (J) Sandstone, Codell Sandstone, Niobrara Formation, while other layers such as Dakota Plainview, Codell Sandstone, Niobrara Formation, Terry and Hygiene sandstone members of the Pierre Shale were discovered later in the 1970's and 1980's. AAPG estimates original reserves to be 1.1 TCFG for the Muddy (J) Sandstone. Fifty years after discovery, the field is currently at peak production because of recent horizontal drilling activity in the Codell and Niobrara with cumulative production to date at 1,060 MMBO and 9.4 TCFG from more than 40,000 wells. Estimated ultimate recovery is 1,100 MMBO and 9.7 TCFG. This represents reserve growth of over 1,000% since the initial discovery.

Geomechanical Setting

The formation was in a generally strike-slip environment, with a very high East-West horizontal stress – i.e., sitting just in front of the Rocky Mountains. The J-Sand was also naturally under pressure (reservoir pressure of about 2,700 psi at a depth of 8,000 ft). This was due to a "natural depletion" from the formation outcrop location far to the East in the state of Nebraska. This depletion resulted in a naturally large stress difference between the J-Sand and the surrounding shale formations. Again, a critical geo-mechanical environment, leading to the successful application of massive hydraulic fracturing. In a single shale stress test, the underlying Skull Creek shale was found to have a minimum stress more than 2,000 psi greater than the stress routinely measured in the sand above. In addition, the J-Sand was generally relatively "simple" geologically. That is a simple 3-layer shale-sand-shale. This made reasonable fracture simulations, using the very first (so called "Pseudo 3D") computer models possible.

Early Development

The field was discovered in 1970, and was credited to an Amoco geologist Roger Matuszczak. Originally 5 "discovery wells" were drilled. These all produced small amounts of gas and were shut-in for land acquisition. Subsequently in late 1970 and early 1971 some 100 delineation wells were drilled, with 72 of these producing natural gas. These wells were fractured with small volume, linear gel treatments. These treatments used typically 40 to 50,000 gallons of gelled water (guar gum gel at 40 to 60 lb/1000 gal concentration). Proppant consisted of 50 to 70,000 lbs – 90% 20/40 mesh white sand and 10% 20/40 mesh glass beads at concentrations up to 3 ppg. The wells responded well but were shut-in after short flow periods to await infrastructure development.

Late in 1971 several wells were placed on a long-term production test and it was a major surprise that the production declined rapidly; and then the wells died due to liquid loading after only a few months. It was the industry's first introduction to transient flow behavior for a fractured well in a low permeability formation (on the order of 0.005 mD in this case). Subsequently several attempts were made using much larger volumes of fluid/proppant and using the early cross-linked gel systems. More than 2/3 of these treatments screened out very early, and the production was no better than that seen for the smaller volume linear gel fracs. Concurrently a new fluid was then developed (proposed by Othar Kiel from Exxon) using an emulsion consisting of 70% condensate and some 30% water and a surfactant. This fluid was then used in several

wells with much better results, but the early screen-out problems persisted. This outcome then resulted in a large field research program. Trials consisted of:

1. Larger pad volumes were trialed, and less sand was placed,
2. A return to the more viscous cross-link gels resulted in even less proppant placed.
3. Higher rates also resulted in less proppant placement, and
4. Fluid loss additives had no effect.

All-in-all, the research resulted in moving from successfully placing 300,000 lbs of sand to an inability to place 100,000 lbs ! Finally, a "pillar frac" was proposed. This was never published/patented but was likely proposed by a man named John Tinsley with Halliburton. This consisted of a pad, followed by proppant laden stages ramping up to 4 ppg, followed by a stage of clean fluid, followed by a stage of 4 ppg, etc. Up to 40 stages of alternating clean fluid/4 ppg were pumped. A typical treatment pump schedule can be seen in [Table 1](#). The "idea" was the sand laden stages would move out into the frac and create "pillars" of proppant with the clean fluid stages leaving lengths of the frac wide open with "infinite" conductivity in between. Clearly idealistic. However, it became possible to place large volume treatments and results substantially improved. A typical "pillar frac" pump schedule is seen in [Table 1](#). ([Fast et al., 1977](#)).

Table 1—"Typical" "Pillar Frac" treatment using condensate emulsion pumped at 30 bpm after (Fast, et al.).

Stage	Content	Name
29,000 gal	Condensate emulsion with no sand	pad
1 stage	4,200 gal with 2 ppg 100 mesh sand followed by 6,300 gal of emulsion with no sand	pillar spacer
1 stage	4,200 gal with 3 ppg 100 mesh sand followed by 6,300 gal of emulsion with no sand	pillar spacer
2 stages	4,200 gal with 4 ppg 100 mesh sand followed by 6,300 gal of emulsion with no sand	pillar spacer
1 stage	6,300 gal with 4 ppg 100 mesh sand followed by 6,300 gal of emulsion with no sand	pillar spacer
8 stages	6,300 gal with 4 ppg 20/40 mesh sand followed by 6,300 gal of emulsion with no sand	pillar spacer
6 stages	8,400 gal with 4 ppg 20/40 mesh sand followed by 6,300 gal of emulsion with no sand	pillar spacer
2 stages	8,400 gal with 4 ppg 10/20 mesh sand followed by 6,300 gal of emulsion with no sand	pillar spacer
1 stage	1,000 gal with 5 ppg 10/20 mesh sand	tail-in
1 stage	6,400 gal with 6 ppg 10/20 mesh sand	tail-in

Following the successful placement of a few large volume treatments, further field tests were conducted in three areas of the field. The goal was to look at the use of larger-and-larger fracture treatments. Summarized results of the tests in two of these areas are included in [Figure 4](#). ([Fast et al., 1977](#)). These two examples represent the best and the worst of the three areas studied. Clearly larger volumes were better in every case. However, since "area C" was most representative of the majority of the field, something better was clearly needed! It is noted that the two 500,000+ gallon treatments were the first two 1,000,000 lb hydraulic fracture treatments ever pumped. For some reason (secrecy on the part of Amoco perhaps), these two treatments never received much attention. The third 1,000,000 lb treatment discussed below was later widely credited as being the first to mark this milestone. Of interest though to modern day frac engineers is the fact that the pump schedules used increasing size proppants including 10/20 mesh sand. This large mesh is rarely used today due to fears of premature screen out, but this was the norm in the heyday of early fracture evolution.

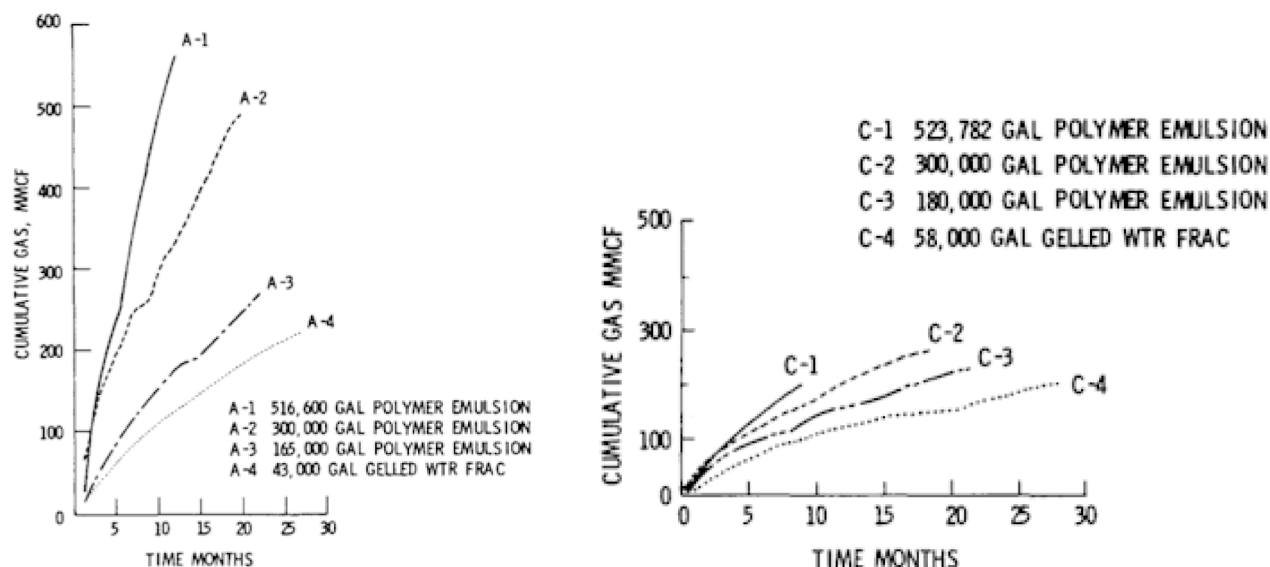


Figure 4—Field test results from small- & large treatments, these represent the best and worst of three areas studied after (Fast, et al.).

Fracturing Pressure Analysis/Redesign

Beginning in 1974, Amoco initiated a research program (led by Robert Fast) into the use of fracturing pressure data to characterize hydraulic fractures. This was eventually published in two distinct papers (Nolte, 1979) and (Smith & Nolte, 1979); with much of the early research done in Wattenberg in 1974 and 75. These papers concentrated on two specific areas: 1) The use of injection pressure data ("net pressure") to characterize fracture geometry including fracture height growth, screenout and natural fracture fluid loss. This included the identification of a "critical" net pressure where undesired fracture geometry changes would occur, and 2) The use of post-injection pressure decline data to identify fluid loss including fluid loss coefficient and the fracturing fluid efficiency. Together, these two analyses could provide the information needed for improved hydraulic fracture treatment designs. These initial pressure analysis techniques have since been expanded by many authors to include the use of the 'G' Function, to determination of the reservoir pressure, and to determination of formation permeability through so called "DFIT" tests.

Unfortunately, all that is left of this past data is seen in the two bottomhole pressure (net pressure) Nolte-Smith plots in Figure 5. One data set of bottomhole pressure was measured for an emulsion "pillar frac" with behavior similar to the lower trace in Figure 5. Similar to this plot, that data clearly showed a geometry change occurring at a net pressure of about 1,700 psi. However, interpretation of this was not definitive. Thus, a second test was planned. Two wells would be used for the pre-frac mini-frac testing followed by measuring bottom hole pressure during the subsequent propped fracture treatment. These wells were actually tested/fractured on the same day in two wells separated by 40+ miles geographically and maybe 600 ft of difference in depth. The behavior was markedly similar for both!

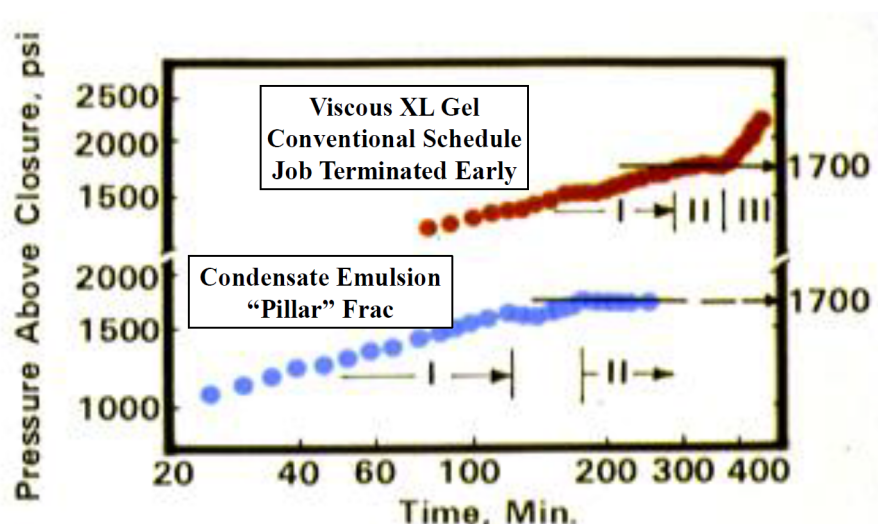


Figure 5—Bottomhole measured net pressure behavior for two Wattenberg frac treatments.

In both cases net pressure initially slowly, steadily increased, indicating a confined height fracture with length extension. Then, at a net pressure of about 1,500 psi, the pressure "bobbled" in both cases. This was theorized to possibly represent the fracture growing upward through the Mowry Shale and being halted by the Huntsman shale. However, there were no in situ stress measurements to substantiate this. There were indications from post-frac temperature logs that there was some vertical fracture height growth out of the J-Sand. In both cases the net pressure then began to increase again prior to leveling off at about 1,700 psi. The upper traces well used a cross-link water at 20 bpm with a "traditional" proppant ramp. In this case the leveling was quickly followed by an increasing pressure trend indicating fracture extension had stopped – i.e., some form of screen-out. The job had to be halted prematurely, and the well was a poor producer. For the lower trace, the well was treated with an emulsion "pillar frac" pumped at 30 bpm (see "typical" pump schedule in Table 1 above). This treatment was pumped to completion and the well was a good producer. That is, the "pillar" design was acting as a pad replenishment to prevent the screenout.

The "flattening" out at 1,700 psi was interpreted to be opening of natural fractures (and the acceleration of fluid loss). This pressure was well below the stress measured in the underlying shale. Also, while height growth could cause the pressure to level – why would height growth cause any form of screen-out? The next permeable formation was 100's of feet shallower. Thus, it seemed that an ideal pump schedule should therefore plan to stay below this critical net pressure of 1,700 psi. That is – less pad, lower rate, and lower viscosity. Totally the opposite of all routine remedies for mitigating treatment screen-out issues. A new design was developed based on these ideas using the first Pseudo-3D fracture simulator. This revised design is compared to the routine "pillar frac" design in Table 2.

Table 2—Revised treatment design, stay below critical net pressure of 1,700 psi

Large "Pillar Frac"	Revised Design
311,000 gallons	186,000 gallons
Condensate Emulsion	Condensate Emulsion
598,000 lbs 24/40 and 10/20 mesh white sand	810,000 lbs 24/40 and 10/20 mesh white sand
10% pad	5% pad
30 bpm	25 bpm reducing to 20 bpm

While no bottomhole pressure was measured, the treatment was pumped with no problems and with no indications of an increasing bottomhole pressure. Over the next 6 months, the well out produced the best of the nearby large volume pillar frac wells by more than 4X. The result on basin development and production is clear from the DJ basin production history in Figure 6.

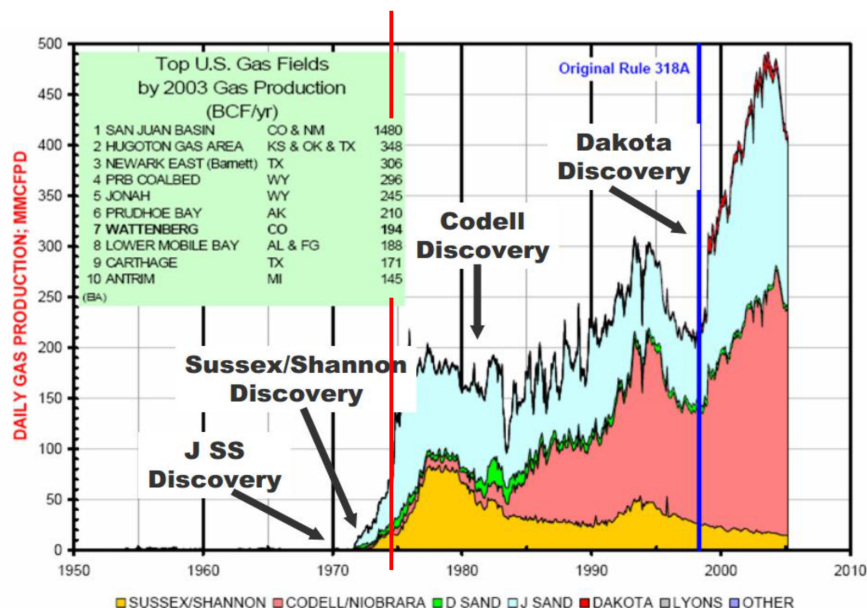


Figure 6—DJ Basin Production history. Date of the Revised Design treatment marked in red.

Following this success, a third 1,000,000 lb frac was then pumped. This was not a significant departure from the "revised design" above, it was just a somewhat thicker formation. While it was technically the third, this was widely hailed as the first 1,000,000 lb frac – probably due to media attention. Within Amoco there was no particular note made of this well – it was just a larger treatment for a thicker formation. However, perhaps the pumping company advertised this as there were several reported photographers on location. This created a great deal of consternation considering the almost 200,000 gallons of highly flammable, raw condensate tanked up on the location. This can be seen in Figure 7 one of the few photographs still existing of the "event".



Figure 7—Picture of first (actually third) 1,000,000 lb fracture treatment, Wattenberg Field, 1975.

Shortly after this the maximum proppant concentration was dropped from 8 to 6 ppg after well tests showed conductivities much higher than required. Routine treatment began to use about 600,000 lbs of sand, and no further treatments larger than this were pumped in the Wattenberg field.

The Fluid

In some ways the "key" to the ideal fluid for the J-Sand was identifiable from the good results with the early, small volume linear gel treatments. At 260 F, these were essentially water fracs. Unfortunately, those key indicators were missed (even when later safety problems prevented further use of the condensate emulsion fluid). In defense of the engineers of that time polyacrylamide friction reducer was not available, and probably no single service company could have marshalled the necessary horsepower needed for a true high rate "water frac". Thus, the condensate emulsion fluid was the "hero". This can be seen from an alphabetical list of the desirable properties for a fluid system.

1. *Breaking*: – So long as the emulsion was being "stirred", i.e., moving, it retained its character with 10's of cP of viscosity. Once it became stagnant, it would then begin to gravity separate as the condensate rose, and the water fell. A reliable breaking mechanism – gravity!
2. *Cheap*: – Fortunately the gas was condensate rich and thus the condensate was readily available. In addition, the operator would charge itself the market price for the base fluid (with no royalties, taxes ... etc. due). Then so long as the well back produced then the basic load of liquids returned, the operator could sell to the market (of course now the government, landowner ... etc. get their cut). The tax advantages were quite significant such that it was a near perpetual motion operation!
3. *Cleanliness*: – There's nothing there but water, condensate, and a surfactant. Thus, "0" residue.
4. *Compatibility*: – The fluid was quite readily compatible with the formation with 70% of the fluid first coming out of the formation itself. This also made the fluid compatible with the inherent reservoir fluids. Finally using minimal water made it easier for this under-pressure reservoir to unload the fracturing fluid. All-in-all, this fracturing fluid was very compatible indeed.
5. *Easy to Mix*: (additive sensitivity) – Again, with only three components this fluid was relatively easy to mix/pump.
6. *Environmentally Friendly*: – Well, pumping a hydrocarbon is never environmentally friendly. However (to our shame) no one worried too much about that 50 years ago.
7. *Fluid Loss*: – There was virtually "0" fluid loss to this tight formation when using this emulsion fluid. From laboratory tests, the fluid loss was on the order of 5X less than the loss exhibited with water.
8. *Viscosity*: – Viscosity is required for three reasons: creating the net pressure and thereby width, preventing fluid loss, and sustaining proppant transport. The emulsion with its moderate viscosity (10's of cP) allowed for the creation of long fractures without exceeding the critical net pressure of 1,700 psi. At the same time, the non-Newtonian structure of the emulsion (like a foam) provided near perfect proppant transport. Finally, as noted above, the fluid loss was near "0". From a viscosity standpoint – all in all an ideal fluid system.
9. *Low Pump Pressure*: – This is the first somewhat negative aspect of the fluid. Due to the loss of hydrostatic head and high friction pressure, this fluid did pump fairly "hard". However, the excellent proppant transport characteristics meant that only low pump rates were needed mitigating this problem.
10. *Safety*: – Again a big negative as seen in [Figure 8](#) Amoco had two major accidents in Wattenberg burning down an orange frac fleet and a red frac fleet. Fortunately, no one was killed (or even hurt) in either accident. In other areas of the country there were a few deadly accidents. After the second accident, in about 1979, the use of the condensate fluid was discontinued. This had a significant impact on post-frac productivity as discussed below.

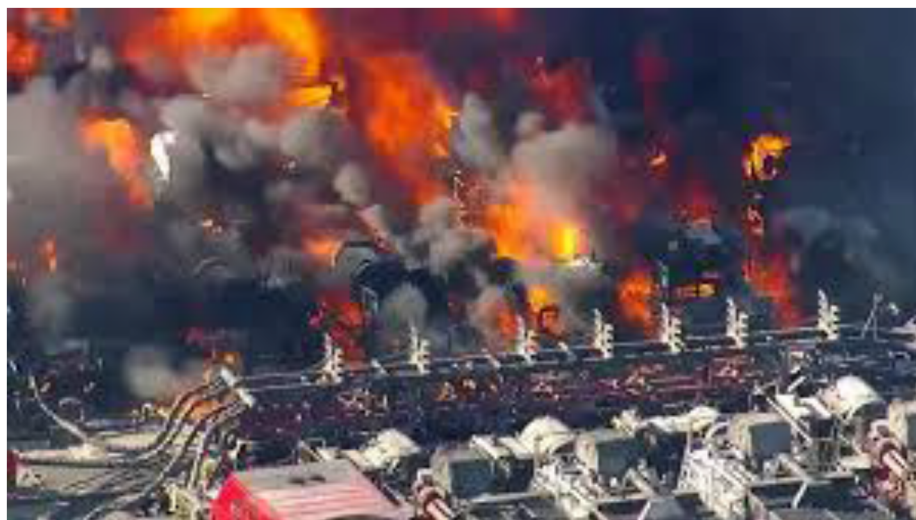


Figure 8—Major safety concerns pumping highly flammable condensate fluid.

For all the reasons outlined above the condensate emulsion was the perfect fluid for the J-Sand. Fortunately, by the time it was abandoned for safety reasons, the cross-linked fluids had improved. Lower gel loadings and the development of some breaker systems made for significant improvements. That, along with smaller pads and lower rates that were used in the early failures still allowed for good results. Despite higher costs and reduced productivity the field development continued. As mentioned earlier the correct option would have been water fracs, however this was not considered at the time. In many respects the J-Sand is the ideal formation for water fracs. It is not too thick (50 to 100 ft) thus leaving much of the formation unpropped is less concerning. Also (pumping water) fracture height confinement is near perfect. Combining this with a relatively high modulus makes for narrow fracture widths and thus, high flow velocity down the fracture – giving an ability to transport sand for long distances. It would be very interesting if a comparison were possible between modern water fracs and the old condensate emulsion fracs. Probably the old fracs would win. On a cost basis they would certainly be cheaper (assuming the tax advantages still exist). Also, from production analysis and long-term shut-in pressure build-up data it was confirmed that effective, propped fracture lengths exceeding 1,500 feet were created. (Roberts, 1981) It is doubtful that this could be achieved with water. Unfortunately, no comparison is possible. By the time water fracs were initiated they were applied to infill wells where depletion existed and wells on the periphery where the limits of the field were being expanded into even poorer rock quality.

Still, on a historical note, it is interesting to look at the use of this fluid to open another tight gas play, the East Texas Cotton Valley (and later Louisiana) formation. The "discovery" well for this development (which eventually expanded in multiple separate fields) used a condensate emulsion fluid ("discovery" in quotes since the formation was a known gas producer, but multiple attempts had failed to achieve commercial production). Circa 1978 Amoco performed the first massive hydraulic fracture treatment in the Jones Gas Unit No.1 in what later became the Woodlawn Field. This treatment used condensate emulsion fluid. However, having to buy the condensate commercially and truck it into East Texas made use of this fluid totally impractical and the fluid was never used again in this formation. Still, this "discovery" well remained the top producer (out of 100's of wells) for at least 20 years!

Conclusions

Wattenberg was the first successful application of large volume (massive) hydraulic fracture treatments. This showed that massive hydraulic fracturing could successfully achieve commercial production from a poor-quality reservoir, with production results far surpassing the three nuclear blast attempts at stimulation.

This success led to further fluid advances and the successful application of this technology to many other formations across the United States.

This success rested on three "legs". First, the formation was an ideal candidate for fracturing with fairly limited thickness and near perfect fracture height confinement. This made the creation of long fractures possible. Second, the ideal fluid system (condensate emulsion) came along just in time, and the condensate rich gas made this fluid system readily useable. Finally, the critical third leg was the development of fracture pressure analysis. This led to understanding the causes of the screen-out problems and allowed the design of a radically different pump schedule. Adding this final "leg" opened up the DJ Basin as an oil and gas province.

This all led to the successful pumping of the first 1,000,000 lb fracture treatments.

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